Numerical Experiments

1 Overview

This document specifies the experimental design, execution, and results used to check consistency between the mathematical model and its implementation and to analyze parameter sensitivity. The model is specified in `1.B.baseline_model/1)baseline/baseline(main).tex`.

The implementation is first verified using a small instance. Further experiments will then use parameter values derived from observations. Sensitivity is examined separately for each parameter.

2 Small Synthetic Instance: Model Verification and Sensitivity Analysis

Purpose. Verify agreement between the formulation and the code using a small synthetic instance that can be checked directly. Vary the inputs to identify ranges of stable optimal allocations, switching points, and changes in throughput and cost.

2.1 Baseline Experiment

2.1.1 Inputs and Common Settings

The values in Table 1 are synthetic inputs chosen for implementation verification. Only market size M_c differs across cities; costs, vehicle productivity, and vehicle limits are identical.

Table 1: Baseline Inputs by City

Parameter	Definition and unit	A	B	C
M_c	Potential trip opportunities (trips/year)	80	100	120
κ_c	Service capacity (trips/vehicle/year)	10	10	10
$\bar{q}_c$	City-specific vehicle limit (vehicles)	2	2	2
F_c	Initial entry cost (USD/city)	20	20	20
K_c	Fixed vehicle cost (USD/vehicle/year)	5	5	5
c_c^+	Unused-capacity penalty (USD/trip)	0	0	0
c_c^-	Forgone contribution margin (USD/trip)	10	10	10

Table 2: Common Settings

Item	Setting
C, Q	$C = \{A, B, C\}$, total fleet limit $Q = 4$ vehicles
S	$\lvert S\rvert = 10$, probability of each scenario $1/\lvert S\rvert = 0.1$
D_c^s	$D_c^s = M_c p_c^s$; p_c^s is a manually specified choice probability (0.05, 0.2, 0.5, 0.8)
Evaluation period	One year after entry: one-time entry cost plus costs and losses for that year
y_c, q_c	y_c, q_c are chosen before observing demand and apply to every scenario
Operations	No vehicle relocation or capacity sharing across cities; demand in cities not entered is also counted as unmet
Verification	SciPy MILP (HiGHS) compared with independent enumeration; no random sampling

The choice probability p_c^s differs from the scenario occurrence probability $1/|S|$. Each row represents one joint demand scenario, not a step in a time sequence. The mean choice probability is 0.38 in every city, and the marginal distributions are identical.

Table 3: Manually Designed Choice Probabilities and Demand Scenarios

s	p_A^s	p_B^s	p_C^s	D_A^s	D_B^s	D_C^s
1	0.05	0.05	0.05	4	5	6
2	0.2	0.2	0.2	16	20	24
3	0.5	0.5	0.5	40	50	60
4	0.8	0.8	0.8	64	80	96
5	0.05	0.2	0.5	4	20	60
6	0.2	0.5	0.05	16	50	6
7	0.5	0.05	0.2	40	5	24
8	0.2	0.5	0.8	16	50	96
9	0.5	0.8	0.2	40	80	24
10	0.8	0.2	0.5	64	20	60
Mean	0.38	0.38	0.38	30.4	38	45.6

2.1.2 Calculation of Performance Metrics

For each city and scenario, served demand, unmet demand, and unused capacity are defined as follows.

$$T_c^s = \min(D_c^s, \kappa_c q_c), \quad U_c^s = \max(D_c^s - \kappa_c q_c, 0), \quad O_c^s = \max(\kappa_c q_c - D_c^s, 0).$$

Averages are arithmetic means over the ten scenarios. **The service rate is calculated across all candidate cities**.

$$\text{SR } (\%) = \frac{\frac{1}{|S|}\sum_{s\in S}\sum_{c\in C} T_c^s}{\frac{1}{|S|}\sum_{s\in S}\sum_{c\in C} D_c^s} \times 100, \qquad \text{CU } (\%) = \frac{\frac{1}{|S|}\sum_{s\in S}\sum_{c\in C} T_c^s}{\sum_{c\in C}\kappa_c q_c} \times 100.$$

A ratio is undefined when its denominator is zero. **Utilization measures the fraction of service capacity used**, not the fraction of time vehicles are operating. The objective and improvement relative to no entry are defined below.

$$\begin{aligned} Z(y,q) &= \sum_{c\in C} F_c y_c + \sum_{c\in C} K_c q_c + \frac{1}{|S|}\sum_{s\in S}\sum_{c\in C}(c_c^+ O_c^s + c_c^- U_c^s), \\ Z_0 &= \frac{1}{|S|}\sum_{s\in S}\sum_{c\in C} c_c^- D_c^s, \\ \Delta Z &= Z_0 - Z(y,q). \end{aligned}$$

The improvement measures the benefit relative to no entry under the same inputs. The objective includes forgone contribution margin and therefore does not equal actual cash expenditure.

Notation used in tables and figures. For $X \in \{D, T, U, O\}$, define the city-level scenario mean $\bar{X}_c = |S|^{-1}\sum_s X_c^s$ and its total $\bar{X} = \sum_c \bar{X}_c$. Thus $\bar{T}$, $\bar{U}$, and $\bar{O}$ denote expected served demand, unmet demand, and unused capacity. SR is the service rate and CU is capacity utilization, both expressed as percentages. For compact cost headers, use

$$C_F = \sum_c F_c y_c, \quad C_K = \sum_c K_c q_c, \quad C_O = \frac{1}{|S|}\sum_{s,c} c_c^+ O_c^s, \quad C_U = \frac{1}{|S|}\sum_{s,c} c_c^- U_c^s.$$

Hence $Z = C_F + C_K + C_O + C_U$ and $\Delta Z = Z_0 - Z$. q^{rep} denotes the representative optimal allocation and Z^* the optimal scenario-based objective. All allocation vectors list cities in the order (A, B, C).

2.1.3 Optimal Allocation and Cost

The optimal solution is $y = (0, 1, 1)$ and $q = (0, 2, 2)$. Independent enumeration of all 60 feasible (y, q) pairs gives the same result, and the baseline optimum is unique. All checks passed: integrality, nonnegativity, city-specific and total fleet limits, demand–capacity balance, and objective recalculation.

Table 4: City-Level Results for the Baseline Optimum (Trips/Year)

c	y_c	q_c	$\kappa_c q_c$	$\bar{D}_c$	$\bar{T}_c$	$\bar{U}_c$	$\bar{O}_c$
A	0	0	0	30.4	0	30.4	0
B	1	2	20	38	17	21	3
C	1	2	20	45.6	17.2	28.4	2.8
Total	2	4	40	114	34.2	79.8	5.8

Table 5: Baseline Cost Breakdown and Performance (USD)

C_F	C_K	C_O	C_U	Z	Z_0	ΔZ
40	20	0	798	858	1,140	282

The service rate is $34.2/114 = 30.0\%$, and capacity utilization is $34.2/40 = 85.5\%$. The capacity of 40 trips is not always fully used. Unused capacity in a low-demand city cannot be transferred to another city, leaving an average of 5.8 trips of unused capacity. Subtracting entry costs of 40 and vehicle costs of 20 from the service contribution margin of $34.2 \times 10 = 342$ gives an improvement of 282.

Table 6: Scenario Totals for Allocation $(0, 2, 2)$ (Trips/Year)

s	$\sum_c D_c^s$	$\sum_c T_c^s$	$\sum_c U_c^s$	$\sum_c O_c^s$
1	15	11	4	29
2	60	40	20	0
3	150	40	110	0
4	240	40	200	0
5	84	40	44	0
6	72	26	46	14
7	69	25	44	15
8	162	40	122	0
9	144	40	104	0
10	144	40	104	0
Total	1,140	342	798	58
Mean	114	34.2	79.8	5.8

2.1.4 Why Concentrate in B and C?

Comparing the six allocations that use the same four vehicles separates the effects of city selection and entry costs. The optimization also considers allocations using zero to three vehicles, not just these six alternatives.

Table 7: Comparison of All Four-Vehicle Allocations (Costs in USD)

q	$\bar{T}$	C_F	C_K	C_U	Z
$(0, 2, 2)$	34.2	40	20	798	858
$(1, 1, 2)$	35	60	20	790	870
$(1, 2, 1)$	35	60	20	790	870
$(2, 0, 2)$	32.8	40	20	812	872
$(2, 2, 0)$	32.6	40	20	814	874
$(2, 1, 1)$	33.8	60	20	802	882

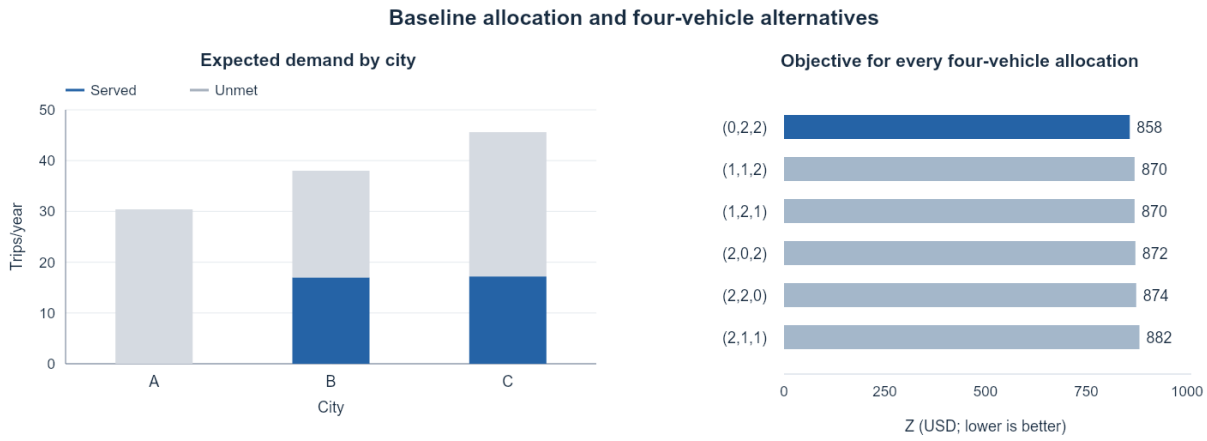


Figure 1: Served and unmet demand by city, and objective values for four-vehicle allocations.

Interpretation. With other city-specific conditions held equal in this instance, selecting the larger markets B and C is advantageous. Capacity limits prevent throughput from increasing in proportion to market size. Allocations $(1, 1, 2)$ and $(1, 2, 1)$ serve 0.8 more trips than $(0, 2, 2)$, but the additional entry cost of 20 exceeds the additional contribution margin of 8, increasing the objective by 12.

2.2 Edge-Case Experiments

Edge cases test whether the model behaves as expected when vehicles or demand are absent, or costs take extreme values. Each experiment changes only the listed input relative to the baseline. Solutions were compared with independent enumeration, and predefined expected properties were checked for all optimal solutions.

Table 8: Modified Inputs and Expected Properties for Edge Cases

Case	Modified input	Expected property
E1	$Q = 0$	$y = q = 0$; $U_c^s = D_c^s$
E2	$M = (0, 0, 0)$	$y = q = 0$; $Z = 0$, $U_c^s = O_c^s = 0$
E3	$\bar{q} = (0, 2, 2)$	$y_A = q_A = 0$
E4	$c^- = (0, 0, 0)$	No service incentive: $y = q = 0$, $Z = 0$
E5	$F = (10000, 10000, 10000)$	Entry costs prohibitively high: $y = q = 0$
E6	$Q = 8$	City limits reached: $q = (2, 2, 2)$, $\sum_c q_c = 6$
E7	$F = (0, 0, 0)$	$q_c > 0 \implies y_c = 1$
E8	$p_c^s = 10/M_c$ $(\forall c, s)$	$D_c^s = 10$; $q = (1, 1, 1)$; $U_c^s = O_c^s = 0$

Table 9: Allocation and Service Results for Edge Cases

Case	q^{rep}	$\bar{T}$	SR (%)	$\bar{U}$	$\bar{O}$	Check
E1	$(0, 0, 0)$	0	0	114	0	PASS
E2	$(0, 0, 0)$	0	—	0	0	PASS
E3	$(0, 2, 2)$	34.2	30	79.8	5.8	PASS
E4	$(0, 0, 0)$	0	0	114	0	PASS
E5	$(0, 0, 0)$	0	0	114	0	PASS
E6	$(2, 2, 2)$	49.8	43.68	64.2	10.2	PASS
E7	$(1, 1, 2)$	35	30.7	79	5	PASS
E8	$(1, 1, 1)$	30	100	0	0	PASS

Table 10: Edge-Case Cost Breakdown and Improvement (USD)

Case	C_F	C_K	C_O	C_U	Z	Z_0	ΔZ
E1	0	0	0	1,140	1,140	1,140	0
E2	0	0	0	0	0	0	0
E3	40	20	0	798	858	1,140	282
E4	0	0	0	0	0	0	0
E5	0	0	0	1,140	1,140	1,140	0
E6	60	30	0	642	732	1,140	408
E7	0	20	0	790	810	1,140	330
E8	60	15	0	0	75	300	225

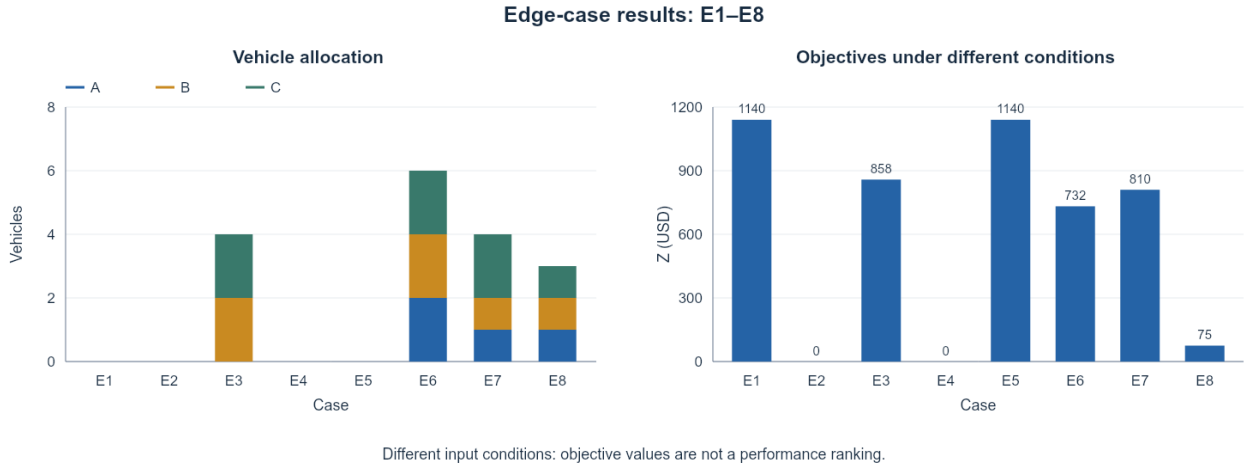


Figure 2: Allocations and objectives for E1–E8. Objectives under different conditions do not constitute a performance ranking.

Interpretation. E1, E4, and E5 all result in no entry, but E4 has a zero objective because the unmet-demand loss coefficient is zero. E6 deploys only six vehicles despite a total limit of eight because of the city-specific limits. E7 has two optimal allocations: $(1, 1, 2)$ and $(1, 2, 1)$. E8 serves all demand with one vehicle per city, leaving the fourth vehicle unused. The service rate in E2 is undefined because demand is zero. Even when $c^+ = c^- = 0$, reported unmet demand and unused capacity were recalculated from the difference between demand and capacity.

2.3 Sensitivity Analysis

2.3.1 Experimental Design and Interpretation

One input type is varied at a time, with all other inputs fixed at their baseline values, and the model is reoptimized. Both allocation changes and performance changes are examined. Experiments that change costs or productivity jointly across cities vary a common level, whereas the M_A experiment changes only A's relative market size.

Table 11: Parameter Ranges for the Main Sensitivity Experiments

Parameter	Variation and range	Fixed inputs
Q	Total fleet limit: 0–8, step 1	$\bar{q}_c = 2$; all other inputs at baseline
F	Common across cities: 0–20, step 1; additional higher values	Vehicle costs, demand, and other inputs at baseline
κ	Common across cities: 5–20, step 0.5	$Q = 4$, $c^+ = 0$; other inputs at baseline
M_A	A only: 0.5–2 times baseline, step 0.05	M_B, M_C, all p_c^s, and other inputs at baseline
c^+	Common across cities: 0–20, step 0.5	Unmet-demand loss, demand, and other inputs at baseline
α	Demand variability multiplier: 0–1, step 0.05	City mean demands and number of scenarios unchanged

Additional tested values of F are $40, 80, 120, 150, 155, 159, 160, 161, 162, 163, 170, 200$. Read the panels in this order: allocation, service rate, cost breakdown, and improvement over no entry. Shaded bands show the range across tied optimal solutions, not confidence intervals. A representative solution is selected by first minimizing the number of entered cities, then using lexicographic order on q and y. Costs can change even when the allocation remains unchanged. In particular, changing M or c^- also changes the comparison benchmark, so the objective alone should not be used to rank performance.

2.3.2 Total Fleet Limit Q

Table 12: Selected Results: Total Fleet Limit (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
0	$(0, 0, 0)$	0	0	1,140	0
3	$(0, 1, 2)$	26.2	22.98	933	207
4	$(0, 2, 2)$	34.2	30	858	282
5	$(1, 2, 2)$	43	37.72	795	345
6	$(2, 2, 2)$	49.8	43.68	732	408
8	$(2, 2, 2)$	49.8	43.68	732	408

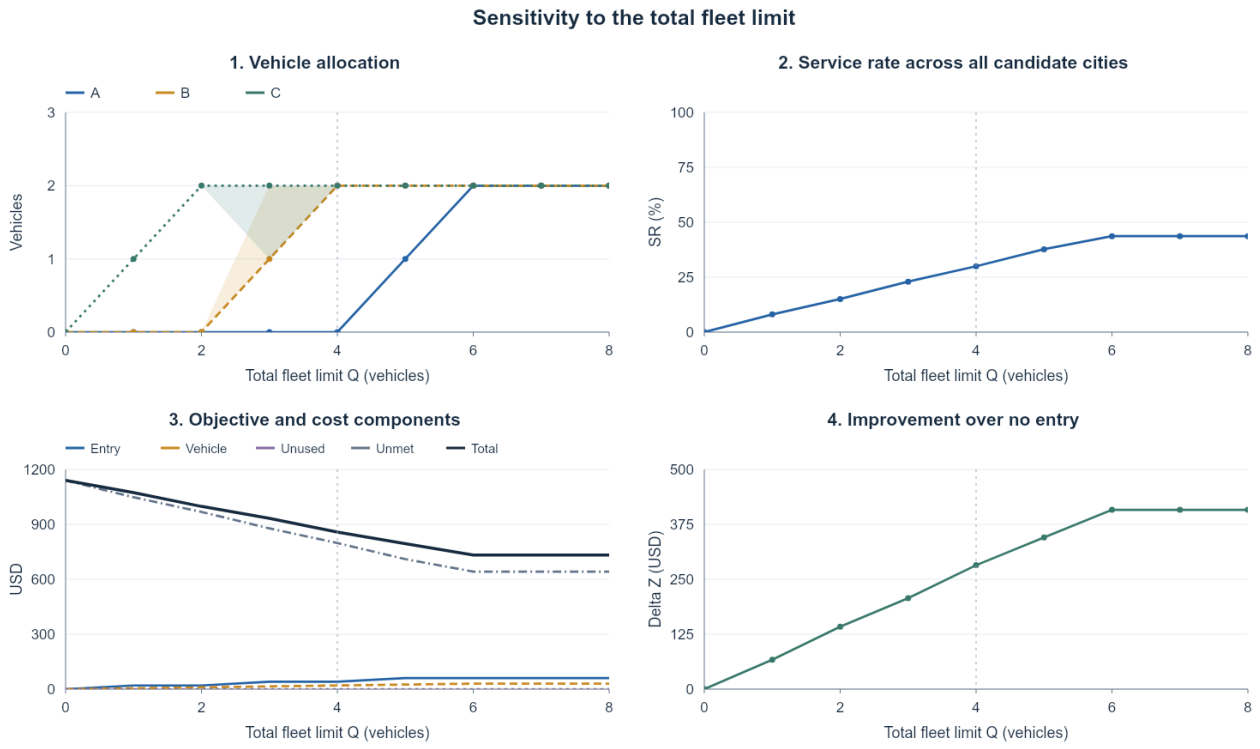


Figure 3: Allocation, service rate, cost, and improvement as the total fleet limit varies.

Interpretation. Increasing Q from 4 to 5 adds one vehicle in A and increases the improvement by 63. Results remain unchanged for $Q \geq 6$ because the city-specific limits sum to six vehicles. This insensitivity reflects deployment limits, not sufficient service of demand. At $Q = 3$, allocations $(0, 1, 2)$ and $(0, 2, 1)$ are tied.

2.3.3 Common Entry Cost F

Table 13: Selected Results: Common Entry Cost (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
0	$(1, 1, 2)$	35	30.7	810	330
8	$(0, 2, 2)$	34.2	30	834	306
20	$(0, 2, 2)$	34.2	30	858	282
160	$(0, 0, 2)$	17.2	15.09	1,138	2
162	$(0, 0, 0)$	0	0	1,140	0
200	$(0, 0, 0)$	0	0	1,140	0

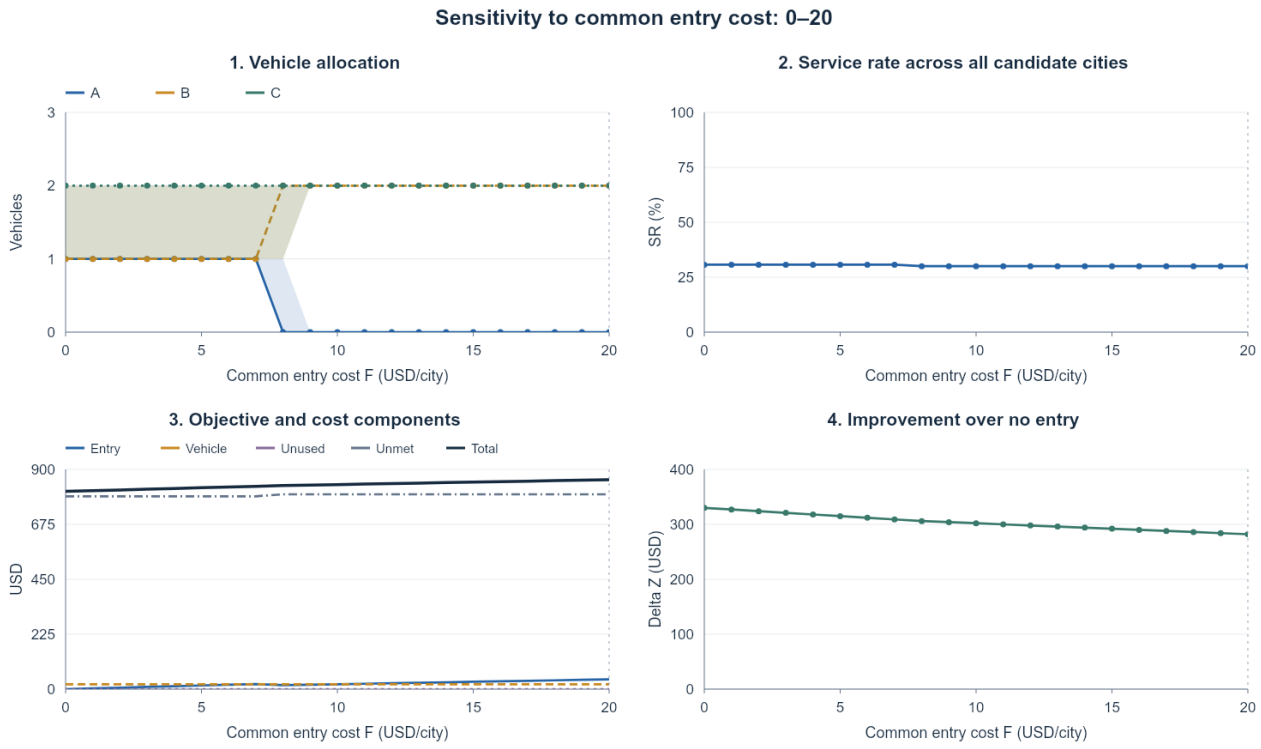


Figure 4: Switch between concentrated and dispersed allocations for entry costs from 0 to 20.

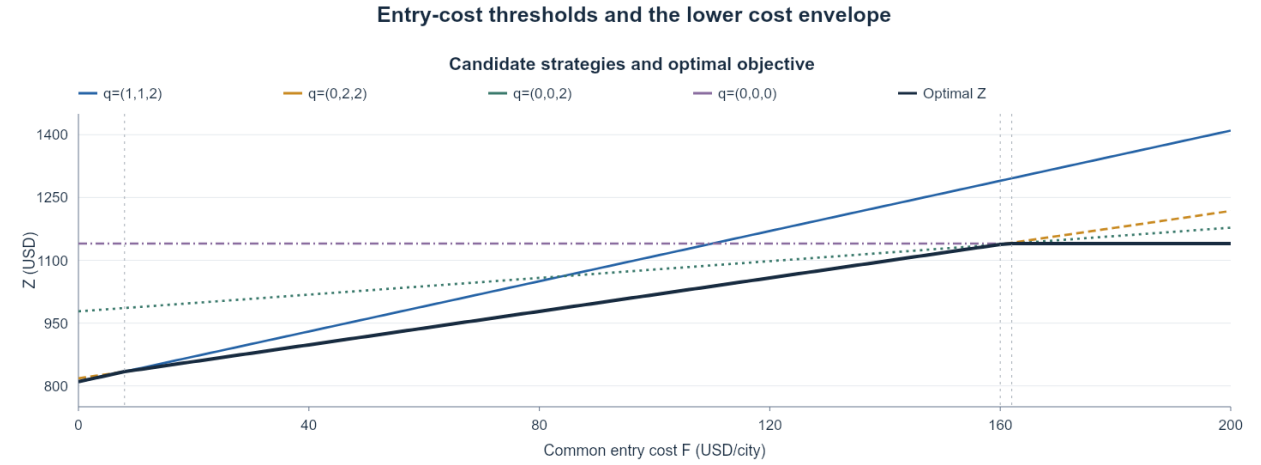


Figure 5: Cost comparison showing switches to single-city entry and no entry at higher entry costs.

Interpretation. When entry costs are low, the additional service from entering all three cities is worthwhile. The two-city allocation costs $818 + 2F$, while the three-city allocation costs $810 + 3F$; they tie at $F = 8$. The lower envelope of all candidate cost functions selects three cities for $F < 8$, B and C for $8 < F < 160$, C alone for $160 < F < 162$, and no entry for $F > 162$. Adjacent strategies tie at the boundaries. These switching points apply to the current synthetic inputs.

2.3.4 Vehicle Productivity κ

Table 14: Selected Results: Vehicle Productivity (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
5	$(0, 2, 2)$	18.2	15.96	1,018	122
10	$(0, 2, 2)$	34.2	30	858	282
12	$(0, 2, 2)$	39.4	34.56	806	334
12.5	$(1, 1, 2)$	42.7	37.46	793	347
20	$(1, 1, 2)$	61	53.51	610	530

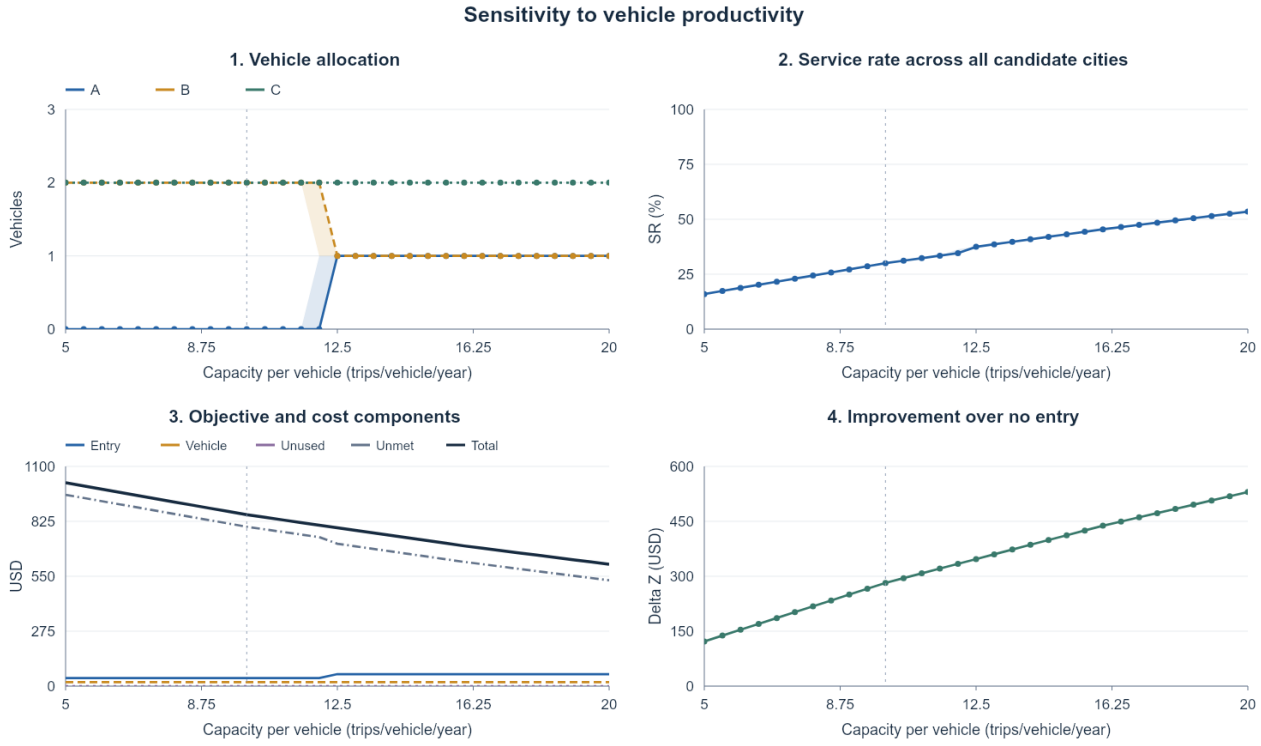


Figure 6: Results from varying service capacity per vehicle.

Interpretation. As productivity increases, the service provided by the first vehicle in A can become more valuable than that provided by the second vehicle in B. Concentrated and dispersed allocations tie at $\kappa = 12$, while a three-city allocation is selected at 12.5. At $\kappa = 20$, the representative allocation is $(1, 1, 2)$, the service rate is approximately 53.5%, and the objective is 610. Unused capacity may increase relative to the baseline while throughput also rises, so unused capacity alone is not a sufficient performance measure.

2.3.5 Market Size of A: M_A

Table 15: Selected Results: A's Market-Size Multiplier (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
0.5	$(0, 2, 2)$	34.2	34.62	706	282
1	$(0, 2, 2)$	34.2	30	858	282
1.25	$(0, 2, 2)$	34.2	28.13	934	282
1.3	$(2, 0, 2)$	34.24	27.81	948.8	282.4
2	$(2, 0, 2)$	34.8	24.1	1,156	288

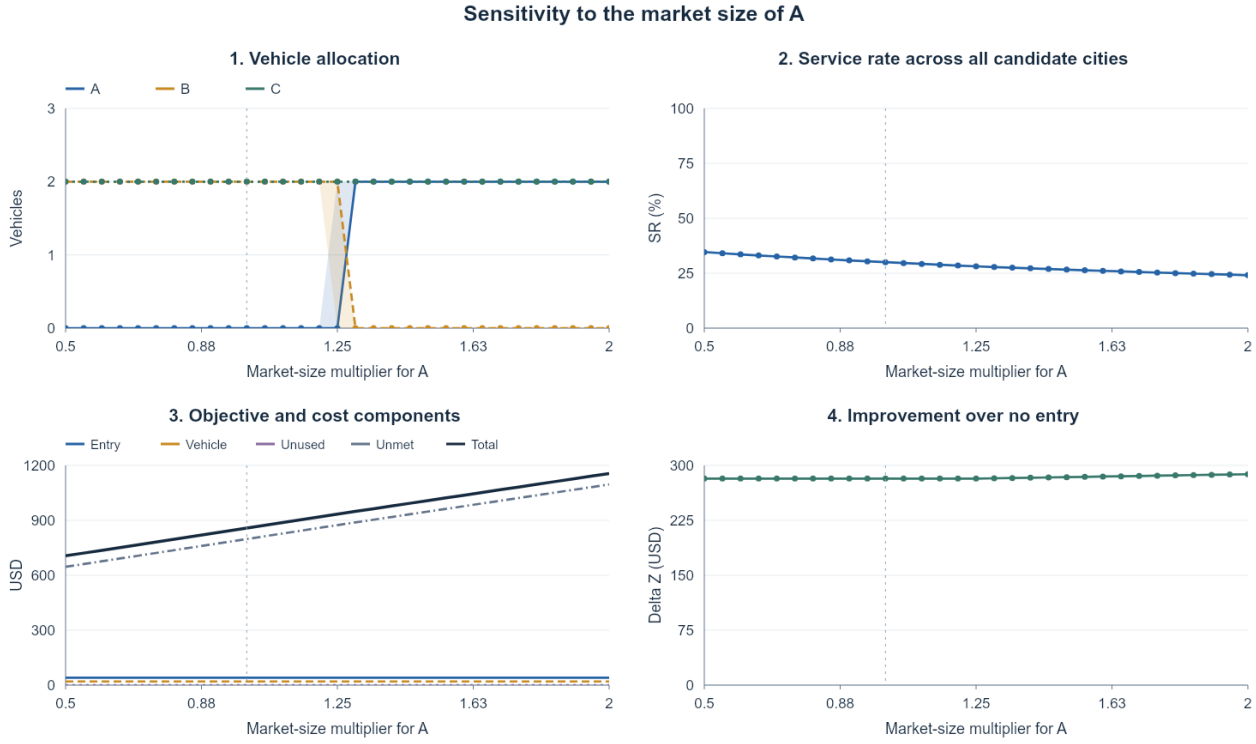


Figure 7: Results from varying only A's market size. A multiplier of 1 corresponds to $M_A = 80$.

Interpretation. At a multiplier of 1.25 ($M_A = 100$), allocations $(0, 2, 2)$ and $(2, 0, 2)$ tie. At 1.30 ($M_A = 104$), allocation $(2, 0, 2)$ is selected, shifting B's two vehicles to A. When A and B have the same market size, their identical marginal choice-probability distributions imply equal expected throughput for the same number of vehicles. Increasing M_A increases both mean demand and the magnitude of demand variation.

2.3.6 Common Unused-Capacity Penalty c^+

Because the baseline value is zero, the coefficient is increased from zero to positive values rather than decreased below zero. This experiment imposes an additional cost on unused capacity beyond the vehicle cost K.

Table 16: Selected Results: Common Unused-Capacity Penalty (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
0	$(0, 2, 2)$	34.2	30	858	282
5	$(0, 2, 2)$	34.2	30	887	253
10	$(0, 2, 2)$	34.2	30	916	224
15	$(0, 2, 2)$	34.2	30	945	195
15.5	$(1, 1, 2)$	35	30.7	947.5	192.5
20	$(1, 1, 2)$	35	30.7	970	170

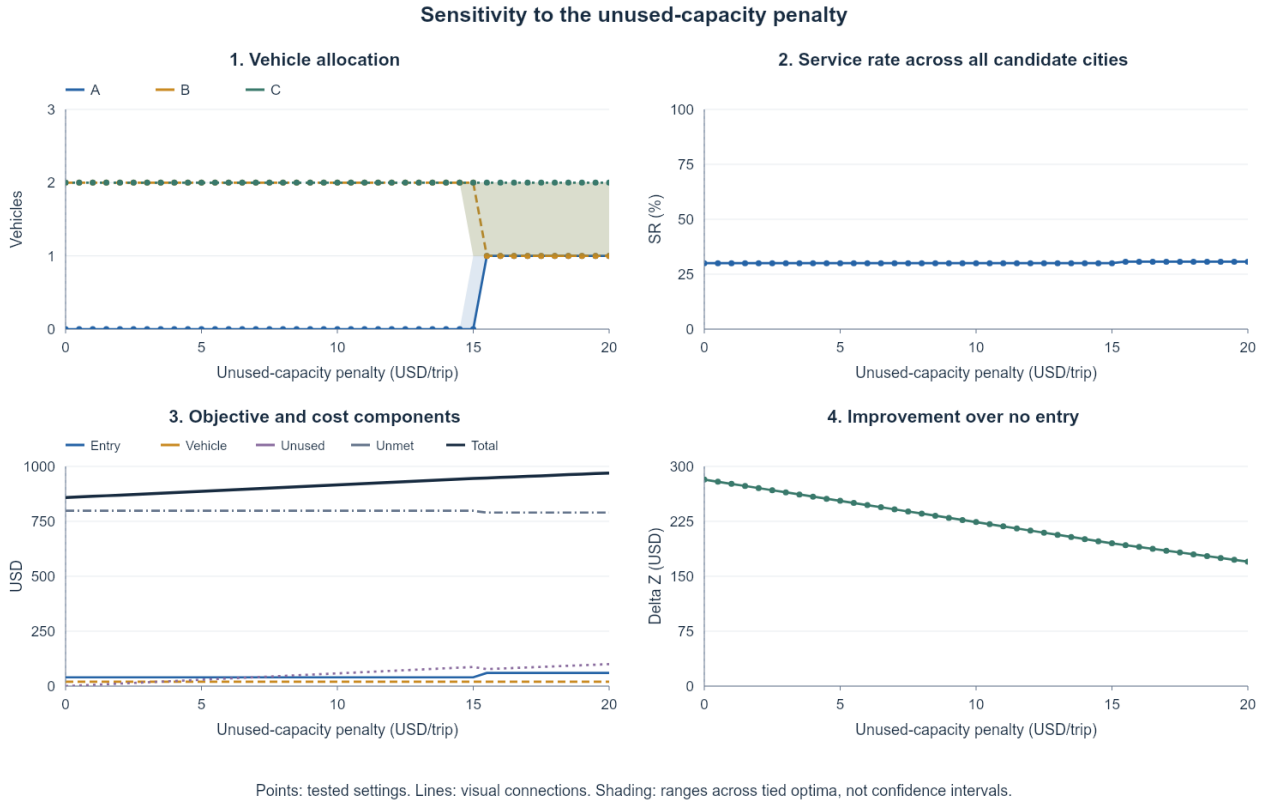


Figure 8: Sensitivity of allocations and performance to the unused-capacity penalty.

Interpretation. At the tested values from 0 to 14.5, allocation $(0, 2, 2)$ remains optimal while its cost increases. Its average unused capacity is 5.8 trips, giving $Z = 858 + 5.8c^+$. The dispersed allocations $(1, 1, 2)$ and $(1, 2, 1)$ have unused capacity of 5 trips and cost $Z = 870 + 5c^+$. The two cost expressions are equal at $c^+ = 15$, where all three allocations tie. At the tested values from 15.5 to 20, the two dispersed allocations are optimal. Following the switch, the service rate increases from 30.0% to approximately 30.70%, but the added unused-capacity cost reduces the improvement at $c^+ = 20$ to 170, below the baseline value of 282.

2.3.7 Demand Variability α with Fixed Mean Demand

$$\bar{p}_c = \frac{1}{|S|}\sum_{s\in S} p_c^s, \qquad p_c^s(\alpha) = \bar{p}_c + \alpha(p_c^s - \bar{p}_c), \quad 0 \le \alpha \le 1.$$

At $\alpha = 0$, every scenario equals mean demand; $\alpha = 1$ recovers the original scenarios. Only the magnitude of variation in the existing scenarios is adjusted; no new random scenarios are sampled.

Table 17: Selected Results: Demand Variability (Throughput in Trips/Year; Costs in USD)

Value	q^{rep}	$\bar{T}$	SR (%)	Z	ΔZ
0	$(0, 2, 2)$	40	35.09	800	340
0.35	$(0, 2, 2)$	40	35.09	800	340
0.4	$(0, 2, 2)$	40	35.09	800	340
0.5	$(0, 2, 2)$	40	35.09	800	340
1	$(0, 2, 2)$	34.2	30	858	282

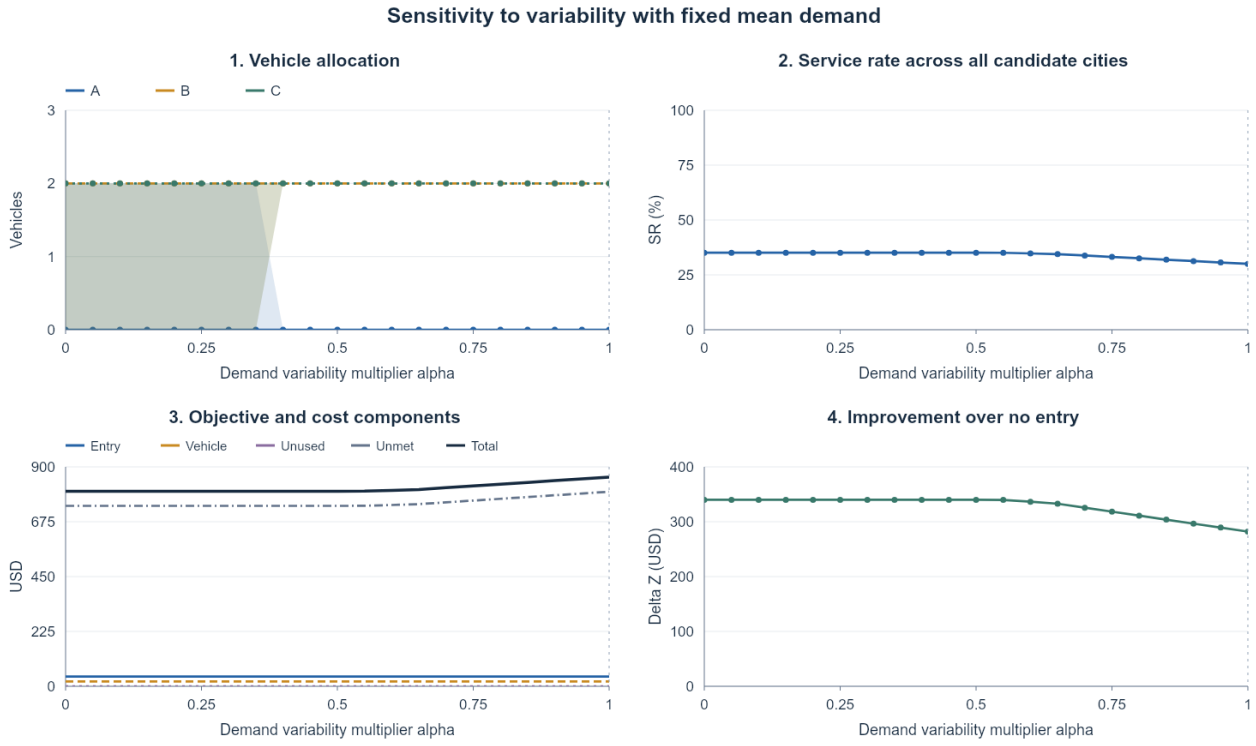


Figure 9: Results from varying demand variability while preserving mean demand.

Interpretation. Without demand variation, allocations $(0, 2, 2)$, $(2, 0, 2)$, and $(2, 2, 0)$ tie. Under the original variability, only $(0, 2, 2)$ is optimal, and the objective increases from 800 to 858. The set of tied optima and performance can change even when the representative allocation is unchanged, so a single representative allocation is insufficient to establish insensitivity.

2.3.8 Three-Level Checks of the Other Main Inputs

Alongside the switching-point analysis, the original three-level experiments for eight inputs are retained. The M multiplier applies to all three cities, as do the other common parameter levels except the system-wide limit Q. Inputs not listed in a row remain at their baseline values. Units follow the input table, except that M is expressed as a multiplier.

Table 18: Three-Level Checks of Eight Inputs: Representative Allocations and Performance

Parameter	Value	q^{rep}	SR (%)	Z	ΔZ
M/M^{base}	0	$(0,0,0)$	—	0	0
M/M^{base}	1	$(0,2,2)$	30	858	282
M/M^{base}	2	$(0,2,2)$	15.96	1,976	304
κ	5	$(0,2,2)$	15.96	1,018	122
κ	10	$(0,2,2)$	30	858	282
κ	20	$(1,1,2)$	53.51	610	530
$\bar{q}$	0	$(0,0,0)$	0	1,140	0
$\bar{q}$	2	$(0,2,2)$	30	858	282
$\bar{q}$	4	$(0,2,2)$	30	858	282
Q	0	$(0,0,0)$	0	1,140	0
Q	4	$(0,2,2)$	30	858	282
Q	8	$(2,2,2)$	43.68	732	408
F	0	$(1,1,2)$	30.7	810	330
F	20	$(0,2,2)$	30	858	282
F	1,000	$(0,0,0)$	0	1,140	0
K	0	$(0,2,2)$	30	838	302
K	5	$(0,2,2)$	30	858	282
K	100	$(0,0,0)$	0	1,140	0
c^+	0	$(0,2,2)$	30	858	282
c^+	2	$(0,2,2)$	30	869.6	270.4
c^+	5	$(0,2,2)$	30	887	253
c^-	0	$(0,0,0)$	0	0	0
c^-	10	$(0,2,2)$	30	858	282
c^-	20	$(0,2,2)$	30	1,656	624

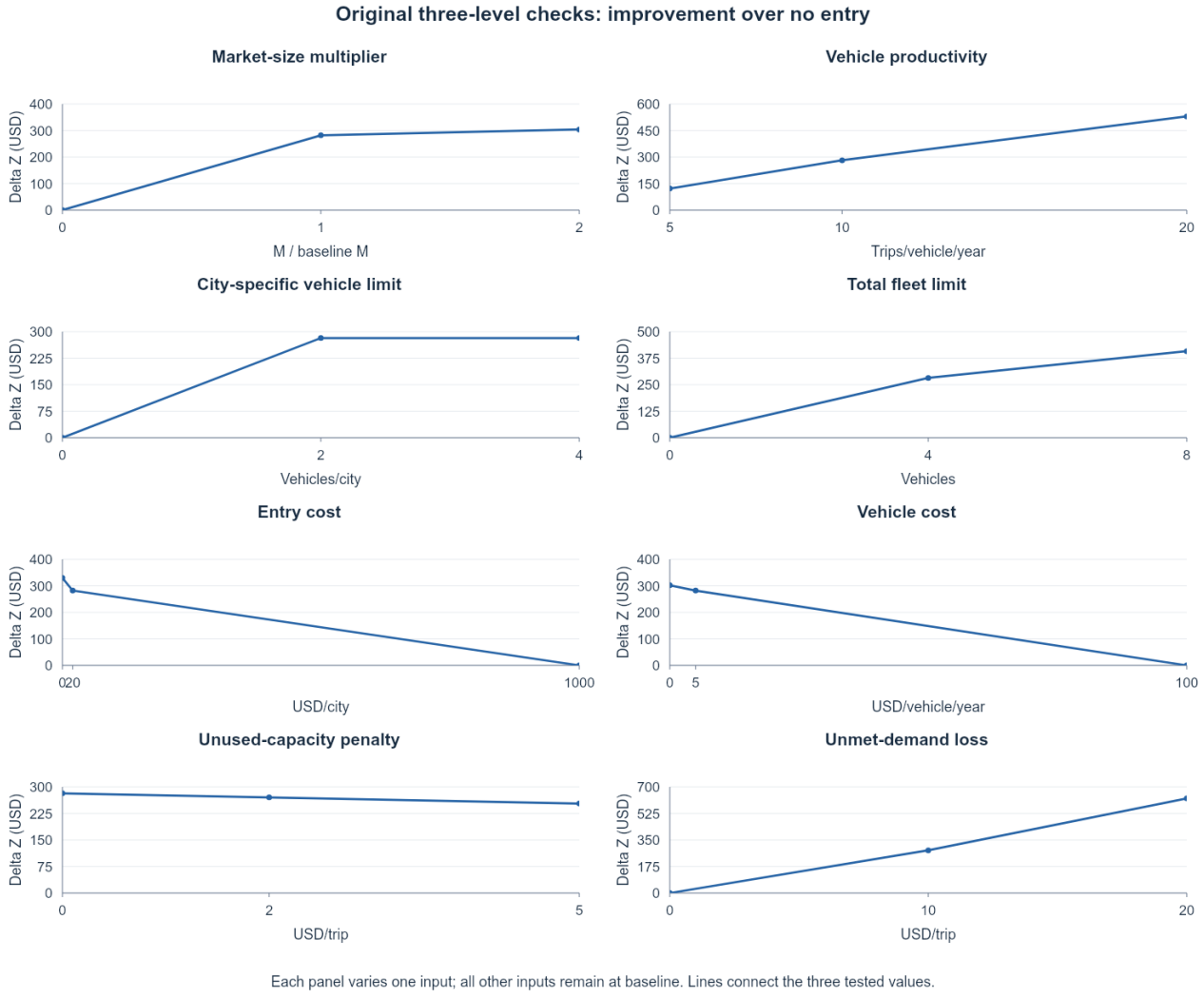


Figure 10: Improvement over no entry for the original 24 settings. This figure alone does not establish exact switching points between the three tested levels.

Interpretation. Doubling M increases the objective, but also increases the improvement from 282 to 304. Relaxing each city's vehicle limit from two to four does not change the optimal cost under baseline conditions. No entry is optimal at $K = 100$. At $c^- = 20$, unmet demand receives a higher monetary valuation even though the allocation is unchanged. Testing only $c^+ = 0, 2, 5$ yields no allocation change, whereas the expanded experiment identifies a switching point at 15. Any claim of insensitivity must therefore state the range examined.